In [6]:
```
# Exp :- 11
```

In [7]:
```
# Aim : To perform Operation of Logistic Regression
```

In [8]:
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:
```

In [9]:
```
import pandas as pd                          # Importing the Libraries
import numpy as np
```

In [10…
```
import os                                     # Data acquisitionuing Pandas
```

In [11…
```
os.getcwd()
```

Out[11]:
```
'C:\\Users\\aryap'
```

In [15…
```
os.chdir("C:\\Users\\aryap\\Downloads")
```

In [16…
```
data=pd.read_csv("heart.csv")
```

In [17…
```
data.head()
```

Out[17]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targ
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	

In [18…
```
data.tail()
```

Out[18]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	ta
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	

In [19…
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       303 non-null    int64  
 1   sex       303 non-null    int64  
 2   cp        303 non-null    int64  
 3   trestbps  303 non-null    int64  
 4   chol      303 non-null    int64  
 5   fbs       303 non-null    int64  
 6   restecg   303 non-null    int64  
 7   thalach   303 non-null    int64  
 8   exang     303 non-null    int64  
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64  
 11  ca        303 non-null    int64  
 12  thal      303 non-null    int64  
 13  target    303 non-null    int64  
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [20…
```
data.describe()
```

Out[20]:

	age	sex	cp	trestbps	chol	fbs	restecg	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	3
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	1
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	1
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	1
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	1
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	2

In [21…
```
data.shape
```

Out[21]:
```
(303, 14)
```

In [22…
```
data.size
```

Out[22]:
```
4242
```

In [23…
```
data.ndim
```

Out[23]:
```
2
```

In [24…
```
data.isna()
```

Out[24]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca
0	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...
298	False	False	False	False	False	False	False	False	False	False	False	False
299	False	False	False	False	False	False	False	False	False	False	False	False
300	False	False	False	False	False	False	False	False	False	False	False	False
301	False	False	False	False	False	False	False	False	False	False	False	False
302	False	False	False	False	False	False	False	False	False	False	False	False

303 rows × 14 columns

In [25…
```
data.isna().any()                    # Data preprocessing _ data cleaning _ m
```

Out[25]:
```
age         False
sex         False
cp          False
trestbps    False
chol        False
fbs         False
restecg     False
thalach     False
exang       False
oldpeak     False
slope       False
ca          False
thal        False
target      False
dtype: bool
```

In [26…
```
data.isna().sum()
```

Out[26]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

In [27…
```
x=data.drop("target", axis=1)   #Independent and Dependent Variables
y=data["target"]
```

In [28…
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2 ,random_
```

In [29…
```
from sklearn.linear_model import LogisticRegression                #Logisti
```

In [30…
```
log = LogisticRegression()
log.fit(x_train, y_train)
```

```
C:\Users\aryap\anaconda3\lib\site-packages\sklearn\linear_model\_logisti
c.py:763: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown i
n:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-re
gression
  n_iter_i = _check_optimize_result(
```

Out[30]:
```
LogisticRegression()
```

In [31…
```
y_pred = log.predict(x_test)
```

In [32…
```
from sklearn.metrics import accuracy_score
```

In [33…
```
accuracy_score (y_test,y_pred)
```

Out[33]:
```
0.8852459016393442
```

In [ ]:

In [ ]: